

Byzantine Robustness Comparison up to $f = 10$: CNN vs. SNN Direct (Complete Sweeps)

ByzFL SNN Benchmark

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1 Introduction

This document presents side-by-side comparison grids (1x2 layout) of Byzantine robustness results comparing Convolutional Neural Networks (CNNs) and Spiking Neural Networks (SNNs) with Direct encoding on MNIST, extended up to $f = 10$ Byzantine clients. The configurations compared are:

1. **CNN**: Convolutional Neural Network on MNIST ('plots_cnn_vs_snn_direct_f10/cnn')
2. **SNN Direct**: Spiking Neural Network with Direct encoding on MNIST ('plots_cnn_vs_snn_direct_f10/snn_direct')

Each grid illustrates the test accuracy under varying numbers of Byzantine clients ($f \in [0, 2, 4, 6, 8, 10]$) and data heterogeneity levels ($\gamma \in [0.0, 0.33, 0.66, 1.0]$) using a single training seed (seed 42), evaluated at the validation-based best step.

2 Best Overall (Worst-Case Across All Attacks)

This section compares performance under the worst-case attack scenario (minimizing the maximum damage over all attacks) for the dynamically selected best aggregator, and for each of the fixed aggregators.

2.1 Best Aggregator Choice

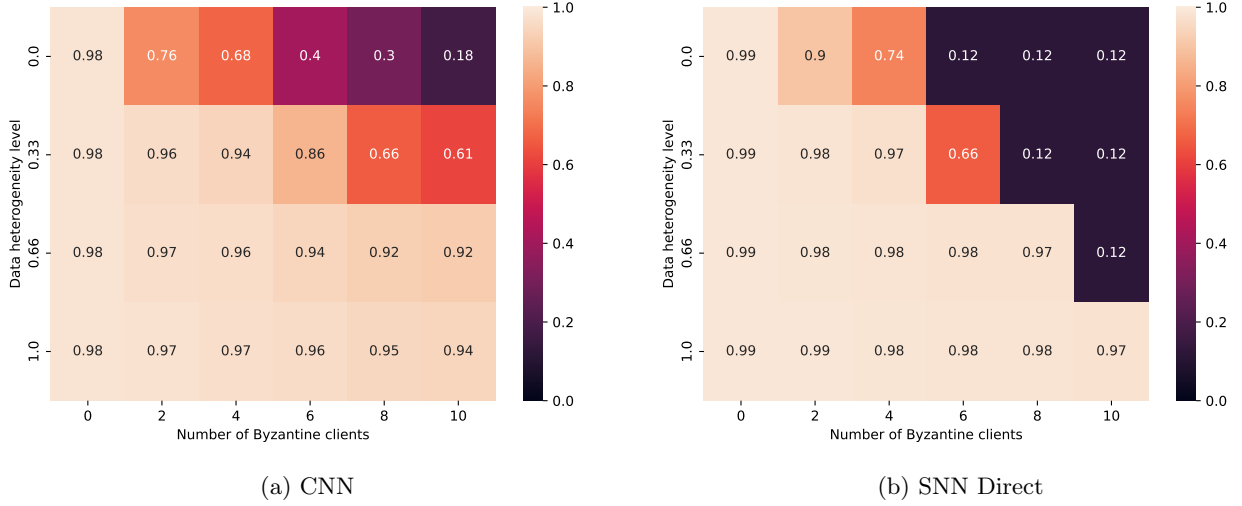


Figure 1: Worst-case test accuracy comparison using the best overall aggregator selection evaluated at the validation-based best step.

2.2 Centered Clipping



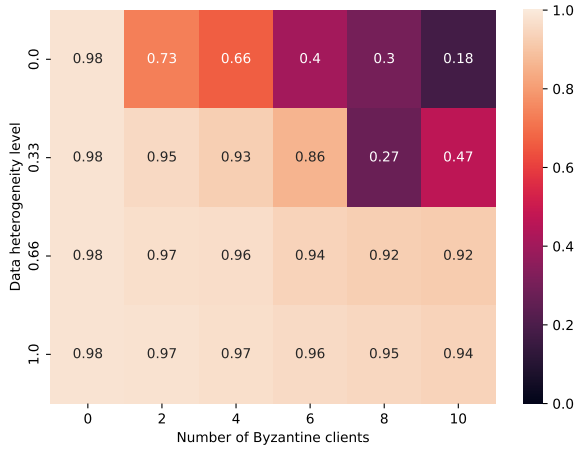
(a) CNN



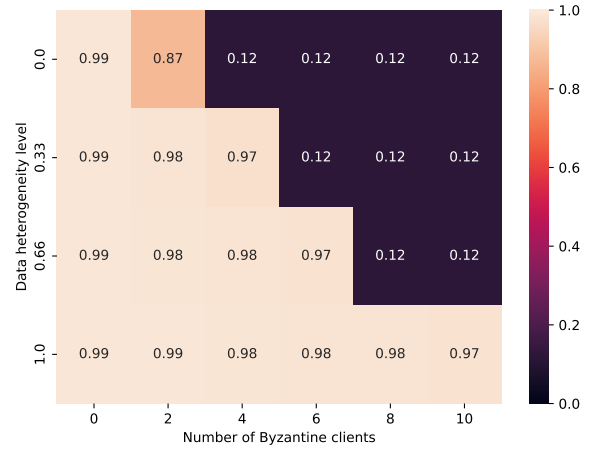
(b) SNN Direct

Figure 2: Worst-case test accuracy comparison using Centered Clipping.

2.3 Geometric Median



(a) CNN



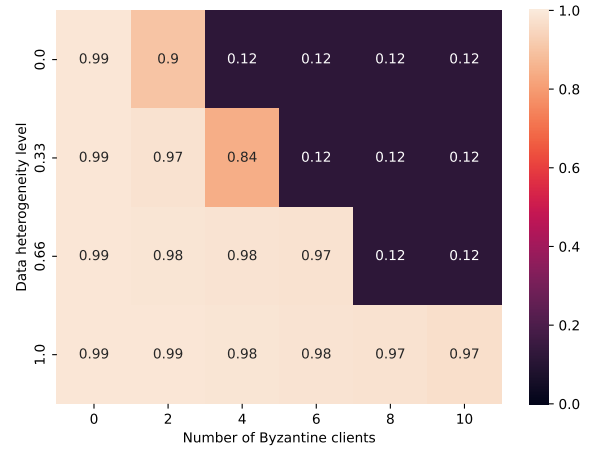
(b) SNN Direct

Figure 3: Worst-case test accuracy comparison using Geometric Median.

2.4 Multi-Krum



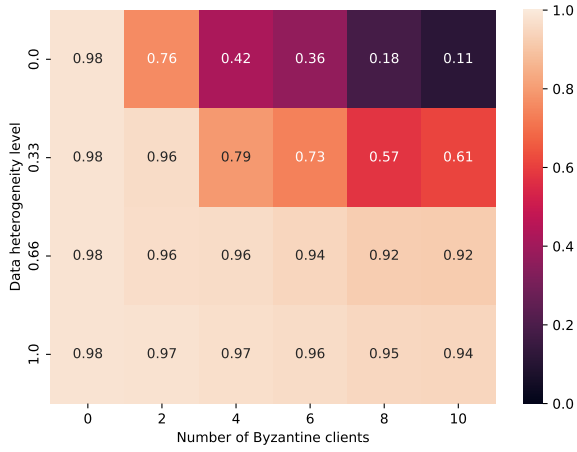
(a) CNN



(b) SNN Direct

Figure 4: Worst-case test accuracy comparison using Multi-Krum.

2.5 Trimmed Mean



(a) CNN



(b) SNN Direct

Figure 5: Worst-case test accuracy comparison using Trimmed Mean.

3 Fixed Attack: Optimal A Little Is Enough (ALIE)

This section compares performance under the original Optimal ALIE Byzantine attack.

3.1 Best Aggregator Choice

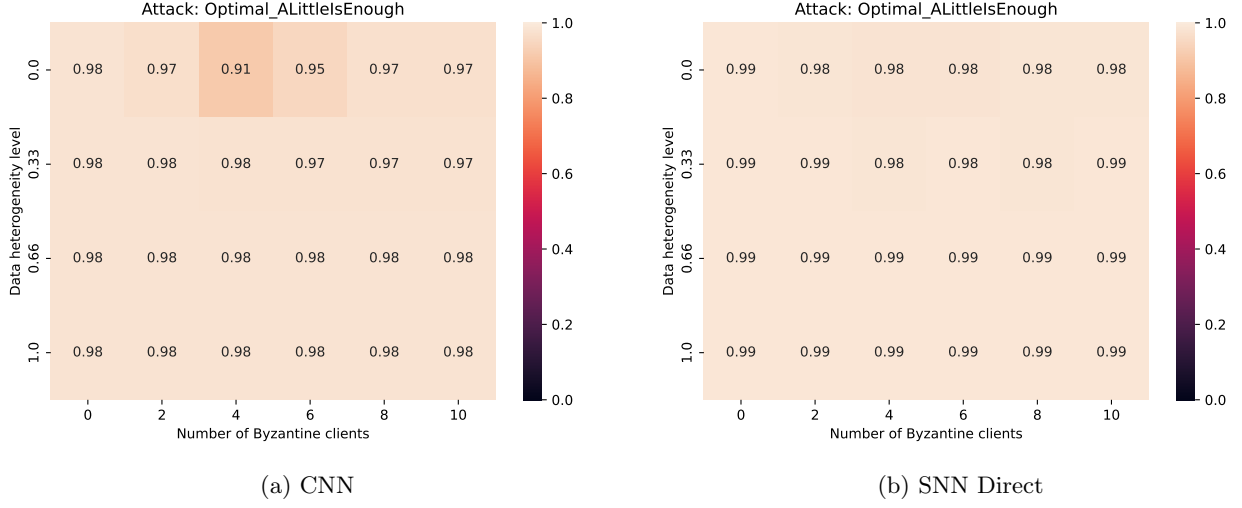


Figure 6: Accuracy comparison under Optimal ALIE attack using the best aggregator selection evaluated at the validation-based best step.

3.2 Centered Clipping

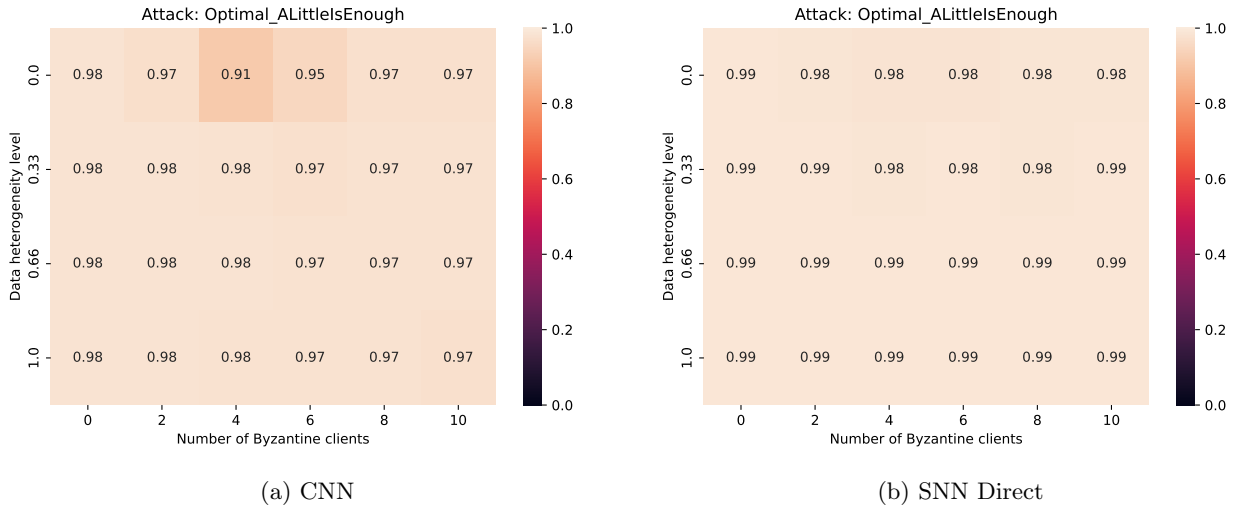


Figure 7: Accuracy comparison under Optimal ALIE attack using Centered Clipping.

3.3 Geometric Median

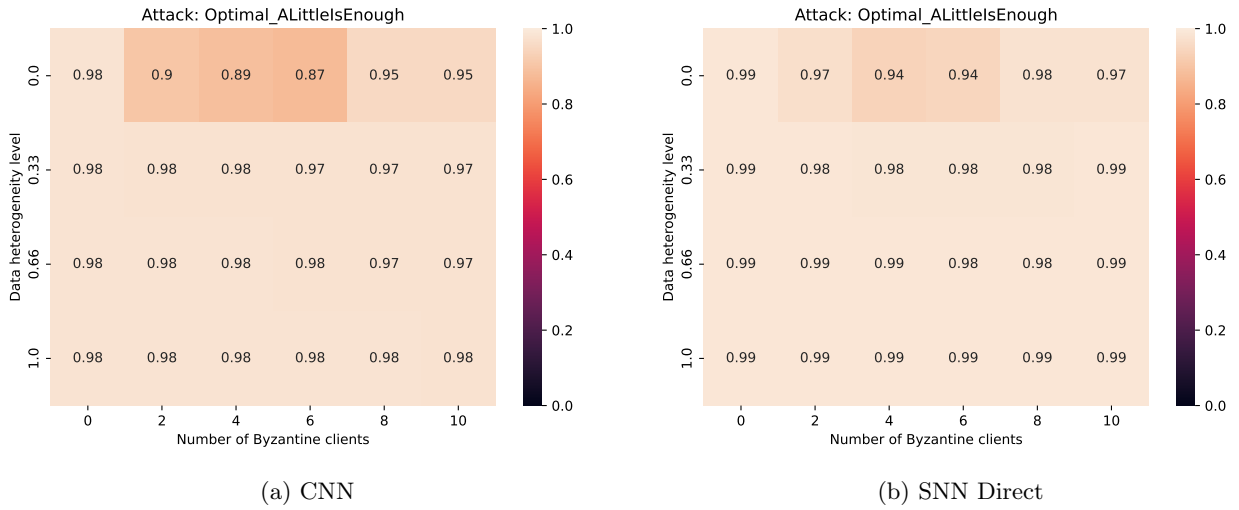
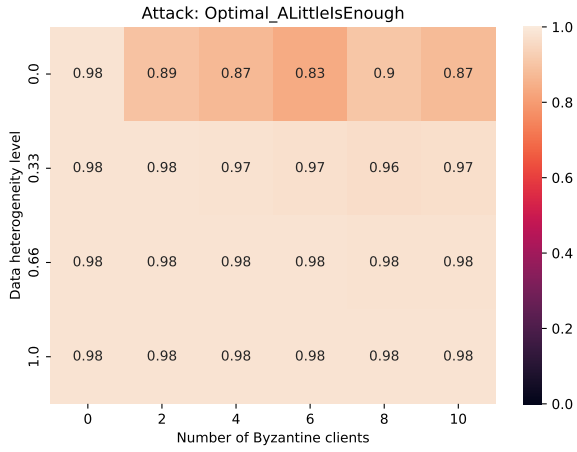
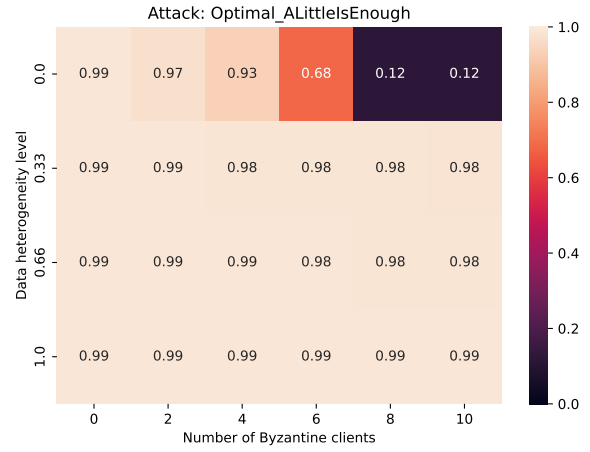


Figure 8: Accuracy comparison under Optimal ALIE attack using Geometric Median.

3.4 Multi-Krum



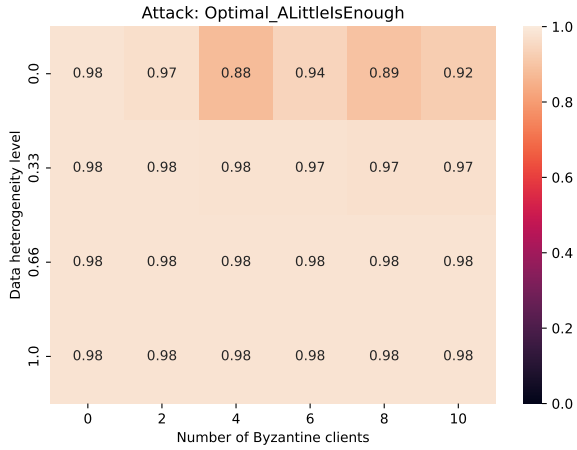
(a) CNN



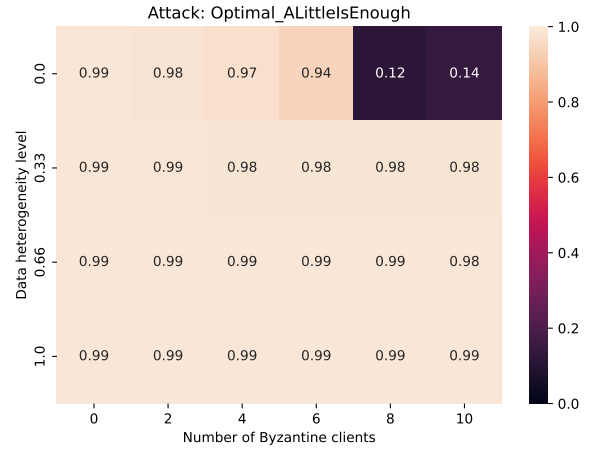
(b) SNN Direct

Figure 9: Accuracy comparison under Optimal ALIE attack using Multi-Krum.

3.5 Trimmed Mean



(a) CNN



(b) SNN Direct

Figure 10: Accuracy comparison under Optimal ALIE attack using Trimmed Mean.

4 Fixed Attack: Optimal Inner Product Manipulation (IPM)

This section compares performance under the Optimal IPM Byzantine attack.

4.1 Best Aggregator Choice

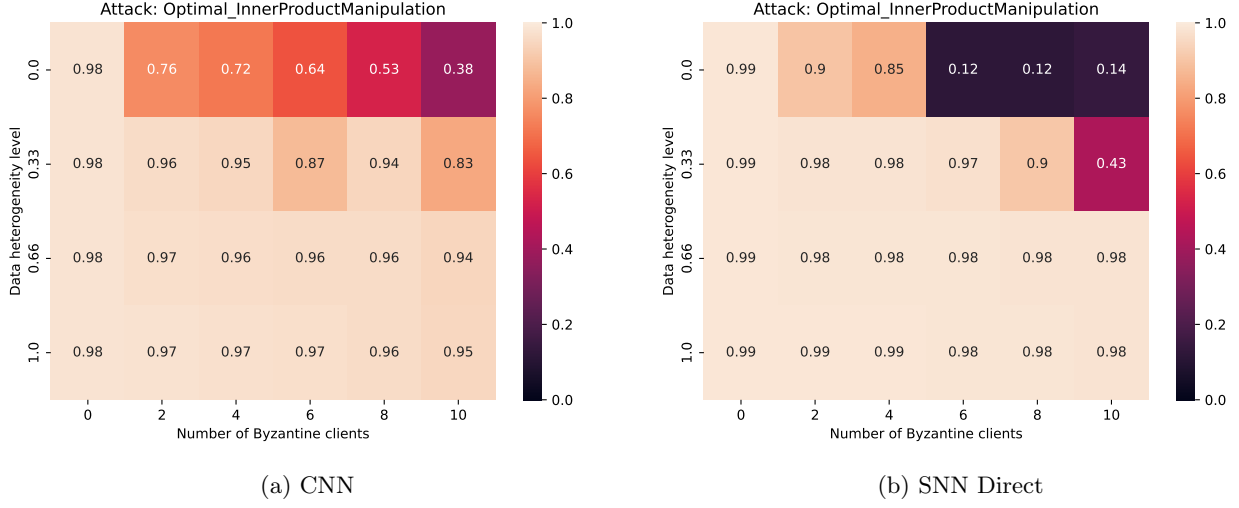
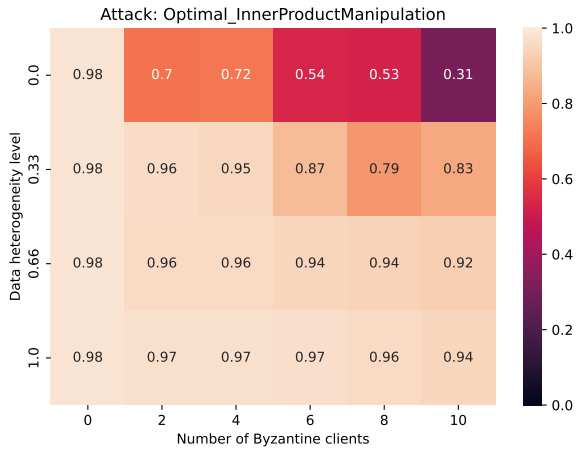
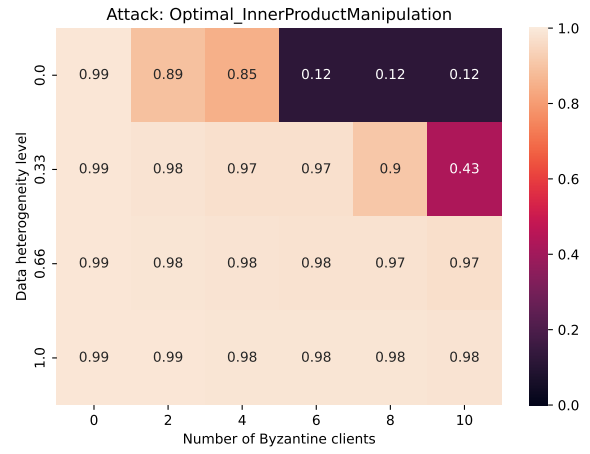


Figure 11: Accuracy comparison under Optimal IPM attack using the best aggregator selection evaluated at the validation-based best step.

4.2 Centered Clipping



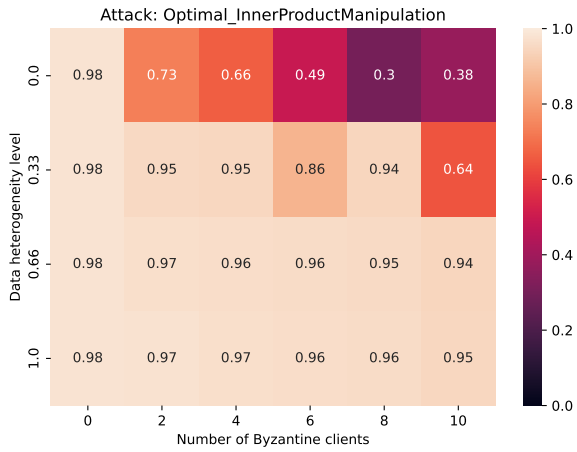
(a) CNN



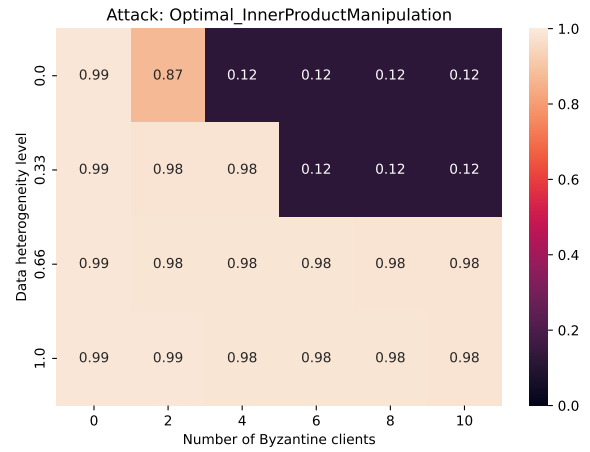
(b) SNN Direct

Figure 12: Accuracy comparison under Optimal IPM attack using Centered Clipping.

4.3 Geometric Median



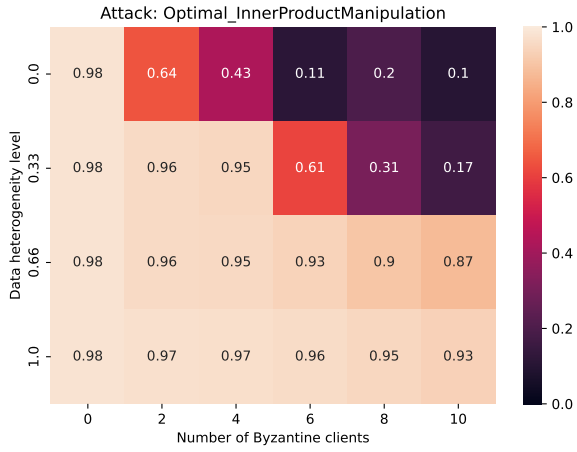
(a) CNN



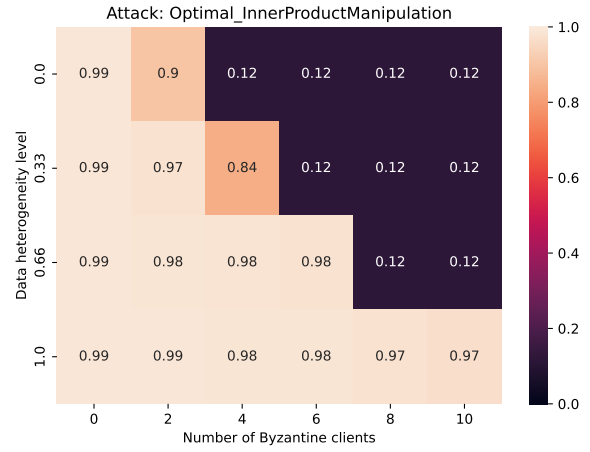
(b) SNN Direct

Figure 13: Accuracy comparison under Optimal IPM attack using Geometric Median.

4.4 Multi-Krum



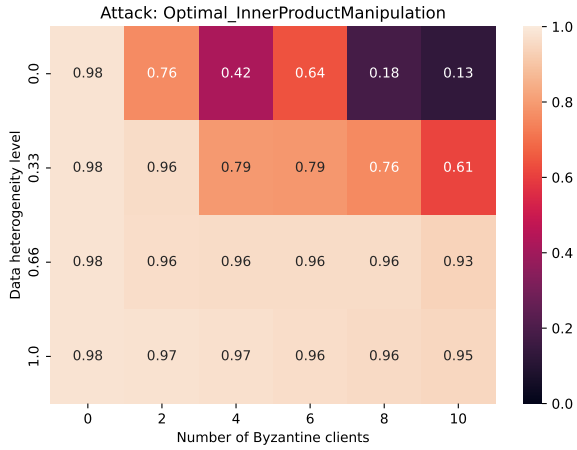
(a) CNN



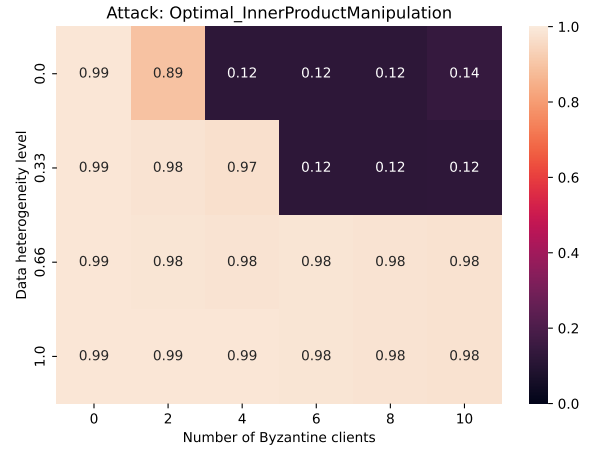
(b) SNN Direct

Figure 14: Accuracy comparison under Optimal IPM attack using Multi-Krum.

4.5 Trimmed Mean



(a) CNN



(b) SNN Direct

Figure 15: Accuracy comparison under Optimal IPM attack using Trimmed Mean.

5 Fixed Attack: Sign Flipping

This section compares performance under the Sign Flipping Byzantine attack.

5.1 Best Aggregator Choice

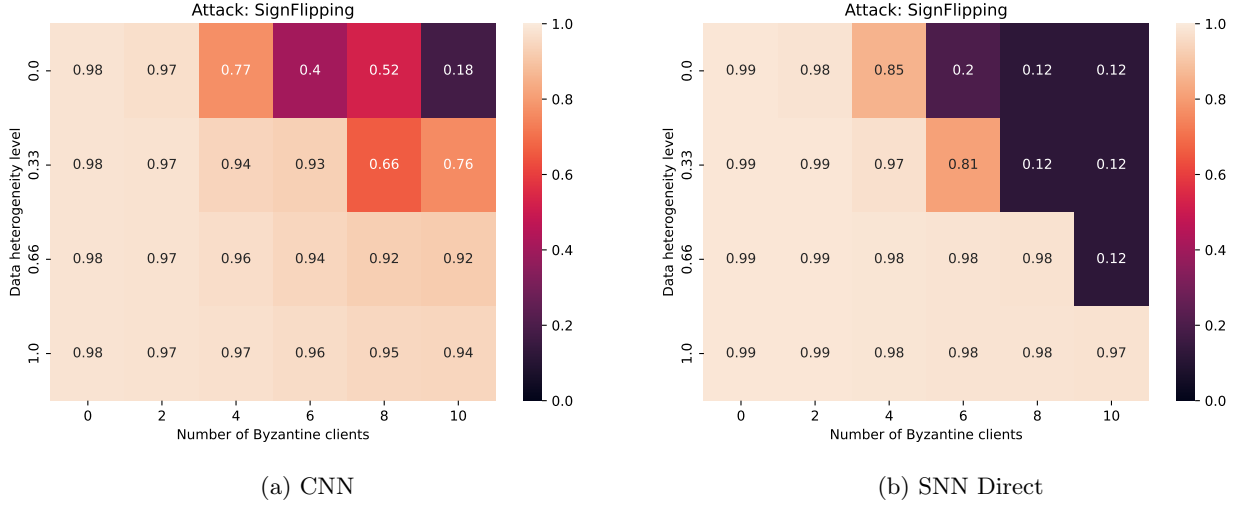
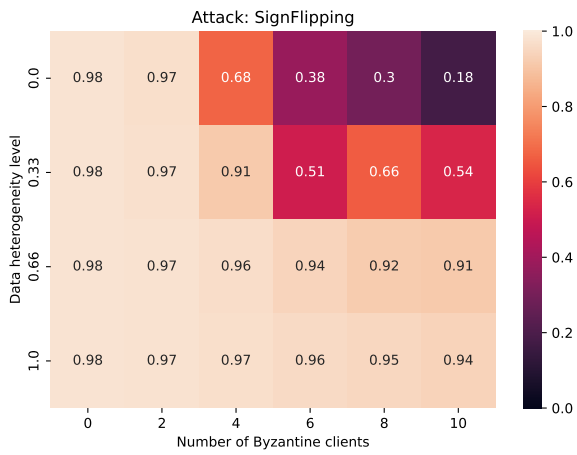


Figure 16: Accuracy comparison under Sign Flipping attack using the best aggregator selection evaluated at the validation-based best step.

5.2 Centered Clipping



(a) CNN



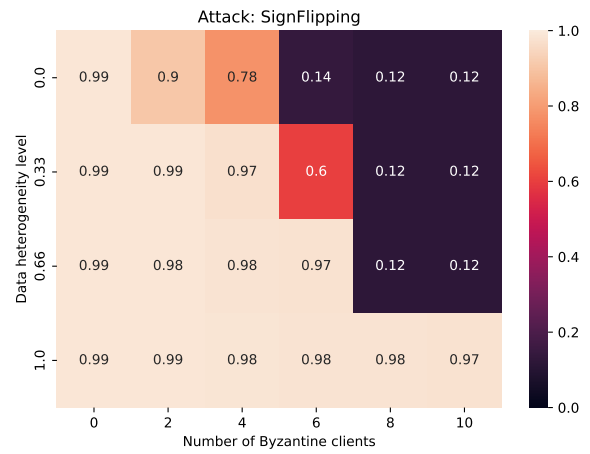
(b) SNN Direct

Figure 17: Accuracy comparison under Sign Flipping attack using Centered Clipping.

5.3 Geometric Median



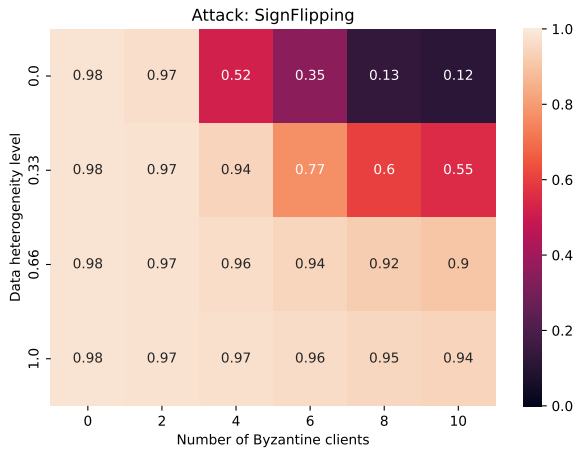
(a) CNN



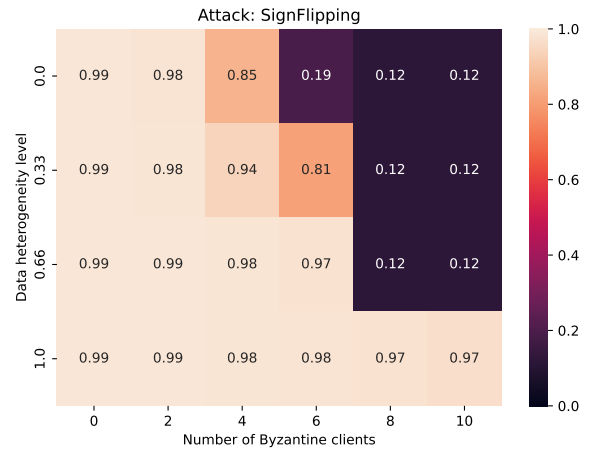
(b) SNN Direct

Figure 18: Accuracy comparison under Sign Flipping attack using Geometric Median.

5.4 Multi-Krum



(a) CNN



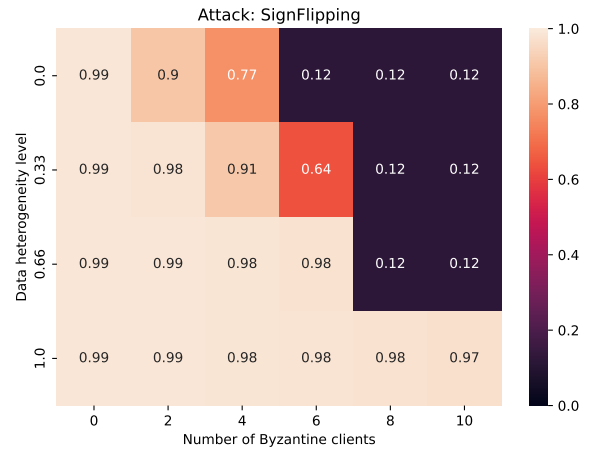
(b) SNN Direct

Figure 19: Accuracy comparison under Sign Flipping attack using Multi-Krum.

5.5 Trimmed Mean



(a) CNN



(b) SNN Direct

Figure 20: Accuracy comparison under Sign Flipping attack using Trimmed Mean.